



基於ESP32 Wi-Fi熱點與網頁控制的動作計數、姿態糾錯和動作調時一體化智能啞鈴

ESP32 Smart Dumbbell with Web Control, Rep Count, Posture Check & Motion Timer

F5C

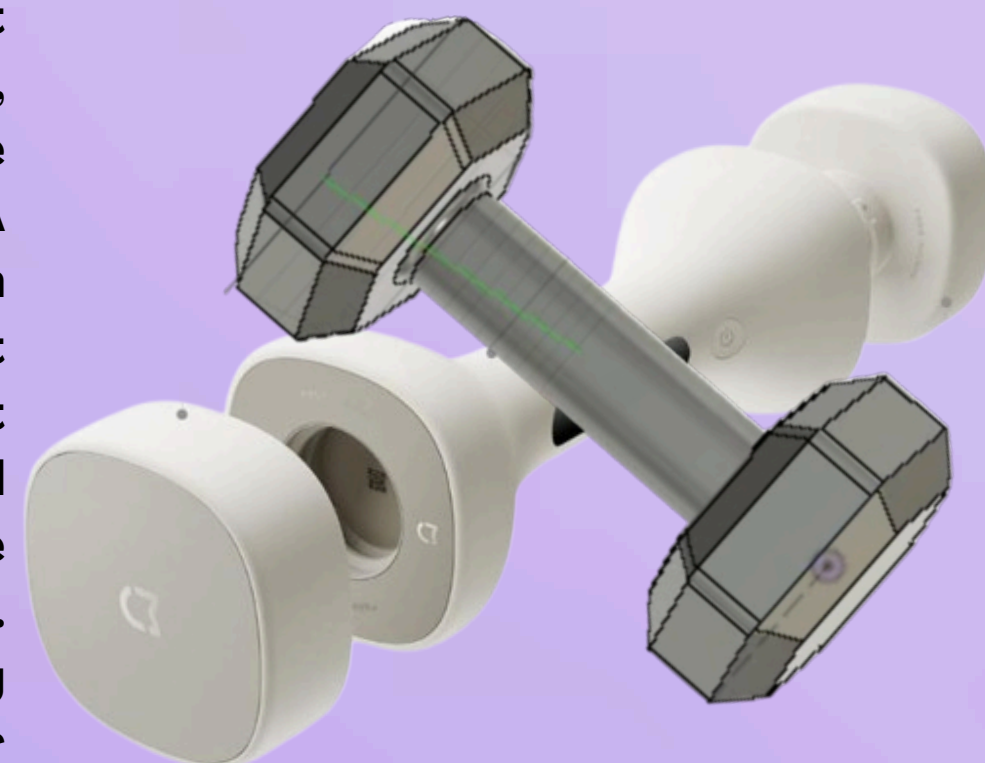
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Smart dumbbell?

The smart dumbbell is designed to help people, particularly the elderly, engage in more exercise by allowing the user to adjust the weight through a connected app or website. In addition, it can detect incorrect movements and provide real-time corrective feedback, which is especially important for older adults who may exercise with improper form and risk injury. This represents a highly promising solution, doesn't it?

So expensive!!!

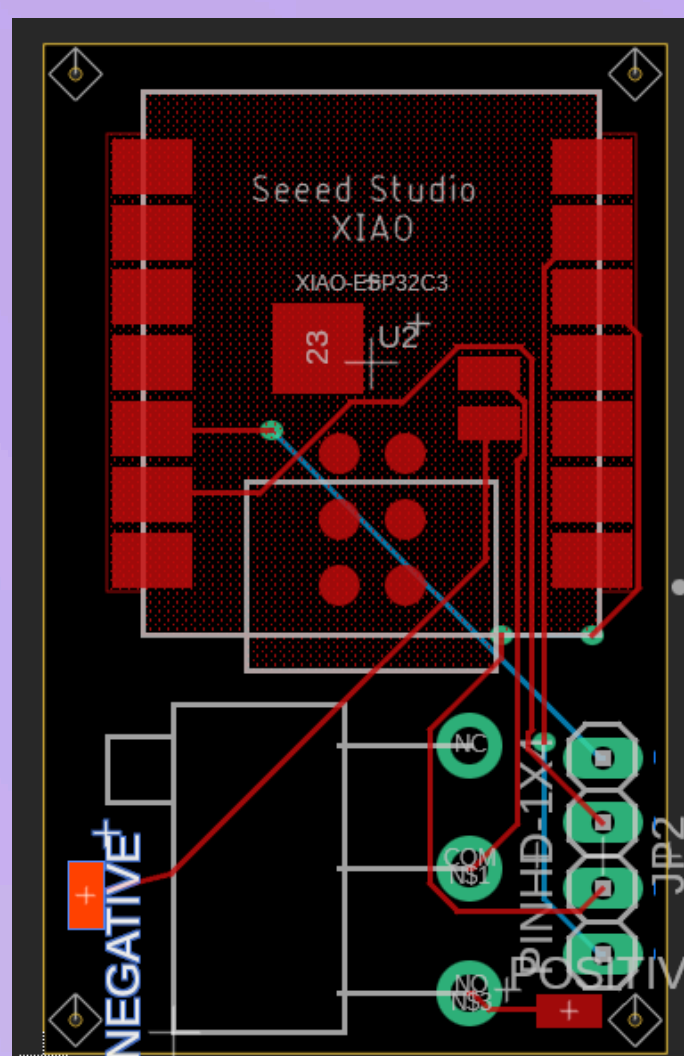
More teenagers are choosing to exercise at home rather than in a gym. In such situations, without proper guidance, they may use equipment incorrectly and risk injury. A smart dumbbell can detect improper form and provide corrective feedback, but it remains relatively expensive. Even the most affordable option from Xiaomi costs around MOP\$90, making it financially inaccessible for many elderly and low-income individuals. As a result, they may continue using traditional dumbbells and face a higher chance of injury.



No Pain, No Gain

During our project, we encountered several challenges, which we addressed one by one. The first issue involved a defective ESP32 Xiao C3 module. After contacting the vendor, they confirmed the fault and provided a replacement, but the time spent troubleshooting and reprogramming caused a significant delay in our schedule. This threatened our ability to meet the deadline; however, by committing additional hours and working overtime, we managed to complete the project on time. Another problem arose with a printed circuit board we had ordered from Mainland China: the shipment became stuck in Zhuhai due to a lost delivery. As a result, we had to place a new order and wait for its arrival, which further impacted our timeline. The third difficulty occurred during the 3D printing of the dumbbell shell. Our first attempt failed because we forgot to include support structures. The second print was then accidentally broken by another student, necessitating a third attempt. This process was extremely time-consuming, as each print required approximately 22 hours.

PCB



What we make

We have designed an affordable, all-in-one smart dumbbell built around a custom-designed circuit board powered by the Sseed Studio XIAO ESP32C3, keeping hardware costs low to ensure accessibility for students and the elderly. The device hosts its own local Wi-Fi hotspot, allowing users to connect directly to a dedicated web control interface from any smartphone or browser. During a workout, the system automatically performs real-time repetition counting, analyzes movement angles to provide instant posture correction, and utilizes a built-in motion timer to regulate exercise pacing. This hardware-software integration delivers automated fitness tracking and immediate safety feedback directly to the user at home, making guided strength training safe and affordable.

FlowChart for the main program



Conclusion

In this project, we successfully designed an affordable, all-in-one smart dumbbell built around a custom circuit board powered by the Sseed Studio XIAO ESP32-C3. By hosting its own local Wi-Fi hotspot, the device allows users to connect directly to a dedicated web control interface from any smartphone browser without needing expensive internet infrastructure. The integrated system successfully delivers real-time repetition counting, posture angle analysis, and exercise pace timing directly to the user at home. Through this development process, we gained extensive hands-on experience in embedded system design, sensor calibration, real-time data processing, and full-stack web development, while learning how to troubleshoot hardware faults and iterate on 3D-printed enclosures to create a reliable, low-cost product.

It finally works, but what will we do next?

Future Outlook and Optimization Plans: Following our initial success, we are now optimizing the hardware enclosure for durability, refining motion algorithms for greater precision, and developing a cloud-based data tracking system to provide users with a complete fitness history and comprehensive progress monitoring.

